

Self-avoiding walks in Anderson localization

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Introduction

The Anderson localization model was introduced by the physicist P.W. Anderson in 1958 [And58] to explain the quantum mechanical effects of disorder in materials. In particular, Anderson's purpose was to study the behavior of electrons in a lattice with disordered potential energy on the sites, which is the quantum mechanical analogue of a random walk in a random environment.

The model is governed by a random Schrödinger operator that describes the total energy of the system. This operator is composed of a kinetic term and a random potential term. The kinetic energy allows the particles to hop from site to site, while the random potential disturbs their motion through the material. When the disorder is sufficiently strong, these particles become trapped in a localized region and can no longer spread to conduct electricity. This phenomenon is known as Anderson localization, and it effectively turns the material into an insulator. This stands in sharp contrast to the behavior of electrons in ideal, perfectly ordered crystals, which naturally act as conductors.

In the last decades, the study of random operators has become an important field of mathematical physics. The two important methods that have been used to prove on which condition on the disorder level Anderson localization holds are the *multiscale analysis* (MSA), developed by Fröhlich and Spencer [FS83], and the *fractional moment method* (FMM) by Aizenman and Molchanov [AM93]. The multiscale analysis method focuses on the bound of the second moments of the operator's resolvent, whereas the fractional moments method utilizes its fractional powers. In this thesis, we are going to focus on the second one, combined with the self-avoiding-walks method.

Structure of the thesis

In Chapter 1, we introduce the Anderson model, more specifically we will distinguish two variations of it: the nearest-sites model and the long-range model. focusing on the properties of its components, its spectrum, and the main results of this thesis.

In Chapter 2, we define spectral and dynamical localization. We also define exponential dynamical localization and show how it implies that all moments of the position operator remain bounded in time.

In Chapter 3, we introduce the Green function as the resolvent of the model and explore how the exponential decay of its fractional power leads to localization. Specifically, we also introduce two central analytical tools, such as the *resolvent identity* of linear operators and the resulting *a priori bound*.

In Chapter 4, we provide the preliminary definitions for self-avoiding walk (SAW) methods and obtain the main results of this thesis using a new representation for the Green function. We will focus on both exponential decay for the standard model, and polynomial decay for the fractional model.

Finally, Chapter 5 is dedicated to the open problems concerning the values of the disorder parameter in different dimensions.

Chapter 1

The Anderson Model

In this chapter, the random Schrödinger operator is the center of our attention. First, we define it as the *Hamiltonian* and discuss the physical meaning of its time-evolving solutions. We are going to define two variations of this model: the standard Anderson model, with near-neighbors jumps, and the fractional Anderson model, with long-range jumps. Right after, we focus on the properties of each of their components and their spectra. Finally, we discuss the main results.

1.1 The Hamiltonian

Transport phenomena in disordered materials are often described via discrete random Schrödinger operators. In quantum mechanics, the *Hamiltonian* represents the total energy of the system, decomposing naturally into kinetic and potential energy. It takes the form of an infinite random matrix on the lattice $\mathbb{Z}^d$,

$$H_{\omega,\alpha} = (-\Delta)^\alpha + \lambda V_\omega, \quad (1.1.1)$$

where $(-\Delta)^\alpha$ is a deterministic matrix representing the kinetic energy, V_ω is a diagonal matrix with random entries representing the disordered potential, and λ is the disorder parameter.

For $\alpha = 1$, the kinetic energy is defined by the standard discrete laplacian, which lets the particles jump from one site to the first-neighbors.

For $\alpha < 1$, the kinetic energy is defined by the fractional discrete laplacian, which provides jumps of any arbitrary distance in the grid.

This operator acts on the Hilbert space of square summable sequences

$$\ell^2(\mathbb{Z}^d) = \left\{ u : \mathbb{Z}^d \rightarrow \mathbb{C} \left| \sum_{x \in \mathbb{Z}^d} |u(x)|^2 < \infty \right. \right\}$$

with the inner product defined as $\langle u, v \rangle = \sum_{x \in \mathbb{Z}^d} \overline{u(x)} v(x)$.

The mathematical space $\ell^2(\mathbb{Z}^d)$ is exactly where the physical reality of the electron is recorded. In quantum mechanics, the state of the system is described by a vector in this space : a normalized wave function $\psi \in \ell^2(\mathbb{Z}^d)$, such that

$$\|\psi\|_{\ell^2}^2 = \sum_{x \in \mathbb{Z}^d} |\psi(x)|^2 = 1.$$

The quantity $P(x) = |\psi(x)|^2$ defines a probability mass function on the lattice $\mathbb{Z}^d$, representing the exact probability of finding the particle at site x .

The state functions are the solutions to the time-dependent Schrödinger equation $H_\omega \psi(t) = i \partial_t \psi(t)$. Because the resulting time evolution operator e^{-itH_ω} is unitary on $\ell^2(\mathbb{Z}^d)$, the total probability is conserved at all times $t \in \mathbb{R}$:

$$\|\psi(t)\|_{\ell^2}^2 = \|e^{-itH_\omega} \psi_0\|_{\ell^2}^2 = \|\psi_0\|_{\ell^2}^2 = 1. \quad (1.1.2)$$

In particular, if the particle is initially placed at a specific site $x \in \mathbb{Z}^d$ with $\psi_0 = \delta_x$, the transition probability to find the particle at site $y \in \mathbb{Z}^d$ at time t is given by the squared transition amplitude:

$$P(x \rightarrow y; t) = |\langle \delta_y, e^{-itH_\omega} \delta_x \rangle|^2. \quad (1.1.3)$$

As we will see later, under certain conditions on the disorder parameter, this probability P tends to 0 at all times t as the distance $|x - y| \rightarrow \infty$, mathematically proving that the particle remains trapped. In particular, the decay changes from exponential to polynomial, as $\alpha = 1$ or $\alpha \in (0, 1)$.

The Discrete Laplacian

The discrete Laplacian is the discrete analogue of the usual Laplacian $(\sum_j \partial^2 / \partial x_j^2)$. First, we consider the properties of this operator for $\alpha = 1$ and then we compare the fractional case, for $\alpha \in (0, 1)$.

$(-\Delta)^\alpha$ for $\alpha = 1$

For the functions $u \in \ell^2(\mathbb{Z}^d)$ it is defined as

$$-\Delta u(x) := \sum_{y \in \mathbb{Z}^d, |x-y|=1} u(x) - u(y) \quad \text{for } x \in \mathbb{Z}^d. \quad (1.1.4)$$

where $|\cdot|$ denotes the euclidean norm.

For each site x , there are $2d$ possible y such that $|x - y| = 1$. Thus, we can define an infinite matrix such that

$$(-\Delta)(x, y) = \begin{cases} 2d & \text{if } x = y \\ -1 & \text{if } |x - y| = 1 \\ 0 & \text{otherwise} \end{cases}$$

The discrete Laplacian represents the kinetic term and connects neighboring sites on the lattice, allowing the electron to transition from one to another.

The operator $-\Delta$ is bounded and self-adjoint on $\ell^2(\mathbb{Z}^d)$. To establish boundedness, we verify **summability** for any $u \in \ell^2(\mathbb{Z}^d)$. We can show this through the following,

$$\begin{aligned} \sum_{x \in \mathbb{Z}^d} |-\Delta u(x)|^2 &= \sum_{x \in \mathbb{Z}^d} \left| \sum_{\substack{y \in \mathbb{Z}^d \\ |x-y|=1}} (u(x) - u(y)) \right|^2 \\ &\leq \sum_{x \in \mathbb{Z}^d} \left(2d \sum_{|x-y|=1} |u(x) - u(y)|^2 \right) \\ &\leq 2d \sum_{x \in \mathbb{Z}^d} \left(\sum_{|x-y|=1} (|u(x)|^2 + |u(y)|^2 + 2|u(x)u(y)|) \right) \\ &\leq 2d \sum_{x \in \mathbb{Z}^d} \left(2 \sum_{|x-y|=1} (|u(x)|^2 + |u(y)|^2) \right) \\ &= 2d \cdot 2 \cdot 4d \|u\|_{\ell^2}^2 = 16d^2 \|u\|_{\ell^2}^2 < \infty, \end{aligned}$$

where we applied the Cauchy-Schwarz inequality to the sum over the $2d$ neighbors, followed by Young's inequality, and finally the translation invari-

ance of the ℓ^2 norm. This guarantees the operator is **bounded**, since

$$\frac{\|-\Delta u\|_{\ell^2}^2}{\|u\|_{\ell^2}^2} \leq 16d^2 \implies \|-\Delta\| = \sup_{u \neq 0} \frac{\|-\Delta u\|_{\ell^2}}{\|u\|_{\ell^2}} \leq 4d.$$

To prove **self-adjointness** for a bounded operator, we only need to show symmetry. For $u, v \in \ell^2(\mathbb{Z}^d)$:

$$\begin{aligned} \langle u, -\Delta v \rangle &= \sum_{x \in \mathbb{Z}^d} \overline{u(x)} \left(\sum_{|x-y|=1} (v(x) - v(y)) \right) \\ &= \sum_{x \in \mathbb{Z}^d} \left(2d\overline{u(x)}v(x) - \sum_{|x-y|=1} \overline{u(x)}v(y) \right) \\ &= \sum_{x \in \mathbb{Z}^d} 2d\overline{u(x)}v(x) - \sum_{x \in \mathbb{Z}^d} \sum_{|x-y|=1} \overline{u(y)}v(x) \\ &= \sum_{x \in \mathbb{Z}^d} v(x) \left(\sum_{|x-y|=1} (\overline{u(x)} - \overline{u(y)}) \right) \\ &= \sum_{x \in \mathbb{Z}^d} v(x) \overline{(-\Delta u(x))} = \langle -\Delta u, v \rangle. \end{aligned}$$

The spectrum of a bounded, self-adjoint operator is a compact subset of $\mathbb{R}$, which in this case is exactly the closed interval $[0, 4d]$. This range represents the allowed kinetic energy levels an electron can possess.

To understand this result explicitly, we apply the discrete Fourier transform:

$$\mathcal{F}: \ell^2(\mathbb{Z}^d) \longrightarrow L^2([0, 2\pi)^d)$$

defined by

$$(\mathcal{F}u)(k) := \lim_{N \rightarrow \infty} (2\pi)^{-\frac{d}{2}} \sum_{|x| \leq N} u(x) e^{-ik \cdot x}, \quad k \in [0, 2\pi)^d.$$

We first assume $u \in \ell^1(\mathbb{Z}^d)$. Because $\ell^1(\mathbb{Z}^d)$ is a dense subset of $\ell^2(\mathbb{Z}^d)$ and the Fourier transform $\mathcal{F}$ is unitary, we can extend the results by con-

tinuity to all functions in $\ell^2(\mathbb{Z}^d)$.

$$\begin{aligned}
(\mathcal{F}(-\Delta u))(k) &= (2\pi)^{-\frac{d}{2}} \sum_{x \in \mathbb{Z}^d} (-\Delta u)(x) e^{-ik \cdot x} \\
&= (2\pi)^{-\frac{d}{2}} \sum_{x \in \mathbb{Z}^d} \left(\sum_{|y-x|=1} (u(x) - u(y)) \right) e^{-ik \cdot x} \\
&= (2\pi)^{-\frac{d}{2}} \sum_{x \in \mathbb{Z}^d} \left(2d u(x) - \sum_{|y-x|=1} u(y) \right) e^{-ik \cdot x} \\
&= 2d (\mathcal{F}u)(k) - (2\pi)^{-\frac{d}{2}} \sum_{x \in \mathbb{Z}^d} \sum_{|y-x|=1} u(y) e^{-ik \cdot x} \\
&= 2d (\mathcal{F}u)(k) - (2\pi)^{-\frac{d}{2}} \sum_{y \in \mathbb{Z}^d} u(y) \sum_{|m|=1} e^{-ik \cdot (y+m)} \quad (\text{setting } m = x - y) \\
&= 2d (\mathcal{F}u)(k) - \left((2\pi)^{-\frac{d}{2}} \sum_{y \in \mathbb{Z}^d} u(y) e^{-ik \cdot y} \right) \sum_{|m|=1} e^{-ik \cdot m} \\
&= 2d (\mathcal{F}u)(k) - (\mathcal{F}u)(k) \sum_{j=1}^d (e^{-ik_j} + e^{ik_j}) \\
&= \left(2 \sum_{j=1}^d (1 - \cos(k_j)) \right) (\mathcal{F}u)(k).
\end{aligned}$$

Thus, the Laplacian acts as a simple multiplication operator in Fourier space

$$\mathcal{F}(-\Delta)\mathcal{F}^{-1} = 2 \sum_{j=1}^d (1 - \cos(k_j))$$

on $L^2([0, 2\pi)^d)$ in the momentum variable $k = (k_1, \dots, k_d)$.

Remark 1.1.1. The discrete Laplacian has no true eigenvalues in $\ell^2(\mathbb{Z}^d)$. However, similar to the continuum Laplacian, it possesses plane waves as generalized eigenfunctions. For a fixed momentum $k \in [0, 2\pi)^d$, we define:

$$\phi_k(x) := e^{-ix \cdot k}.$$

While $\phi_k \notin \ell^2(\mathbb{Z}^d)$ because it is not square-summable, the operator $(-\Delta)$

acts on it as

$$\begin{aligned}
(-\Delta\phi_k)(x) &= \sum_{|m|=1} (e^{-ix \cdot k} - e^{-i(x+m) \cdot k}) \\
&= 2d\phi_k(x) - \left(\sum_{j=1}^d 2\cos(k_j) \right) \phi_k(x) \\
&= \left(2 \sum_{j=1}^d (1 - \cos(k_j)) \right) \phi_k(x).
\end{aligned}$$

Thus, ϕ_k is a bounded, generalized eigenfunction of $(-\Delta)$ corresponding to the spectral value $2 \sum_{j=1}^d (1 - \cos(k_j))$. Since $\cos(k_j)$ oscillates between -1 and 1 , the spectrum of the discrete Laplacian is purely continuous and strictly bounded within the interval $[0, 4d]$.

$(-\Delta)^\alpha$ for $0 < \alpha < 1$

The fractional kinetic operator is formally defined via the discrete Fourier transform by raising to a fractional power:

$$\mathcal{F}(-\Delta)^\alpha \mathcal{F}^{-1} = \left(2 \sum_{j=1}^d (1 - \cos(k_j)) \right)^\alpha.$$

Like the standard discrete Laplacian, $(-\Delta)^\alpha$ is bounded, self-adjoint, and translation-invariant.

However, its behavior in position space is fundamentally different. Instead of only connecting nearest neighbors, the fractional operator connects arbitrarily distant sites. Consequently, for any $x \neq y$, the matrix elements $(-\Delta)^\alpha(x, y)$ are strictly negative and never exactly zero. Instead, they exhibit a polynomial decay. There exist strictly positive constants $0 < c_{\alpha,d} \leq C_{\alpha,d} < \infty$ such that:

$$\frac{c_{\alpha,d}}{|x - y|^{d+2\alpha}} \leq -(-\Delta)^\alpha(x, y) \leq \frac{C_{\alpha,d}}{|x - y|^{d+2\alpha}} \quad \forall x \neq y. \quad (1.1.5)$$

Because $d + 2\alpha > d$ for any $\alpha \in (0, 1)$, this polynomial decay ensures that the off-diagonal kernel is absolutely **summable**:

$$\sum_{y \neq x \in \mathbb{Z}^d} |(-\Delta)^\alpha(x, y)| \leq C_{\alpha,d} \sum_{y \neq x \in \mathbb{Z}^d} \frac{1}{|x - y|^{d+2\alpha}} < \infty. \quad (1.1.6)$$

This summability is crucial because it guarantees that the diagonal entries are strictly positive, well-defined, and finite:

$$(-\Delta)^\alpha(x, x) = - \sum_{y \neq x} (-\Delta)^\alpha(x, y) > 0 \quad \forall x \in \mathbb{Z}^d. \quad (1.1.7)$$

Remark 1.1.2. The diagonal relation (1.1.7) is the direct fractional analogue of the standard discrete Laplacian, where we had $(-\Delta)(x, x) = - \sum_{|x-y|=1} (-1) = 2d$. The critical distinction is that the standard Laplacian computes this value by summing over exactly $2d$ nearest neighbors. In the fractional case, because the particle can jump to any site, we sum over the entire lattice.

Definition 1.1.3 (Step weight kernel). Let $\alpha \in (0, 1]$ and $s \in (0, 1)$. Then we define

$$D(x, y) := |(-\Delta)^\alpha(x, y)|^s,$$

as the *step weight kernel*.

One can check the summability of the step weight kernel for $\alpha \in (0, 1)$; more precisely, for polynomial decay.

Lemma 1.1.4. For any $\alpha \in (0, 1)$ and $s \in \left(\frac{d}{d+2\alpha}, 1\right)$,

$$\sum_{x \in \mathbb{Z}^d} |(-\Delta)^\alpha(0, x)|^s < \infty \quad (1.1.8)$$

Proof. From (1.1.5):

$$\sum_{|x| \geq 1, x \in \mathbb{Z}^d} |(-\Delta)^\alpha(0, x)|^s \leq C_{\alpha, d}^s \sum_{|x| \geq 1, x \in \mathbb{Z}^d} \frac{1}{|x|^{(d+2\alpha)s}}.$$

By standard comparison of the lattice sum with an integral on $\mathbb{R}^d$:

$$\sum_{|x| \geq 1, x \in \mathbb{Z}^d} \frac{1}{|x|^p} \leq C_d \int_{|z| > 1} \frac{1}{|z|^p} dz.$$

Passing to spherical coordinates in $\mathbb{R}^d$, the integral reads

$$\begin{aligned} \int_{|z| > 1} \frac{1}{|z|^p} dz &= |\mathbb{S}^{d-1}| \int_1^\infty \frac{r^{d-1}}{r^p} dr \\ &= |\mathbb{S}^{d-1}| \int_1^\infty r^{d-1-p} dr, \end{aligned}$$

where $|\mathbb{S}^{d-1}| = \frac{2\pi^{d/2}}{\Gamma(d/2)}$ denotes the surface area of the unit sphere in $\mathbb{R}^d$.

The radial integral converges if and only if the power satisfies

$$d - 1 - p < -1 \iff p > d.$$

Substituting $p = s(d + 2\alpha)$, this condition becomes

$$s(d + 2\alpha) > d \iff s > \frac{d}{d + 2\alpha}.$$

Since $\alpha > 0$ implies $d + 2\alpha > d$; for every $s \in \left(\frac{d}{d+2\alpha}, 1\right]$, we conclude that

$$\sum_{x \neq 0} |(-\Delta)_{0,x}^\alpha|^s \leq C_{\alpha,d}^s \sum_{x \neq 0} \frac{1}{|x|^{s(d+2\alpha)}} < \infty.$$

□

Thus, for $s \in \left(\frac{d}{d+2\alpha}, 1\right)$, $D(x, y)$ generates a transition probability

$$p(x \rightarrow y) := \frac{D(x, y)}{\sum_{w \neq x} D(x, w)}$$

for $x \neq y$ in $\mathbb{Z}^d$, and is translation-invariant.

Remark 1.1.5. For $\alpha = 1$ the step weight kernel takes the form $D(x, y) = (2d)^s \delta_{x=y} + \delta_{|x-y|=1}$.

Because the transition probability excludes the diagonal $w = x$, the $2d$ term vanishes from the transitions, and the denominator evaluates simply to the coordination number of the lattice

$$\sum_{w \neq x} D(x, w) = \sum_{w: |x-w|=1} 1 = 2d.$$

Thus, the transition probability perfectly gives a simple random walk on $\mathbb{Z}^d$

$$p(x \rightarrow y) = \begin{cases} \frac{1}{2d} & \text{if } |x - y| = 1, \\ 0 & \text{otherwise.} \end{cases}$$

and is well-defined for all $s \in (0, 1)$.

The Random Potential

The random potential is expressed as the multiplication operator $V_\omega(x) = \omega_x$. It is a diagonal matrix, where $\omega = (\omega_x)_{x \in \mathbb{Z}^d}$ is a set of independent identically distributed real-valued random entries. Physically, V_ω represents the random electric potential created by nuclei at the sites $x \in \mathbb{Z}^d$.

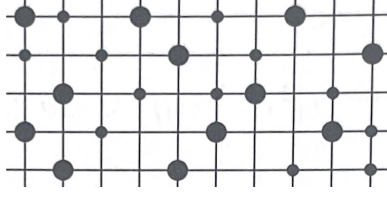


Figure 1.1: A disordered lattice system. Picture from [AW15b]

Definition 1.1.6 (i.i.d. random variables). Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a general probability space. A family of random variables $(\omega_x)_{x \in \mathbb{Z}^d}$ defined on $(\Omega, \mathcal{A}, \mathbb{P})$ is **identically distributed** if there exists a Borel probability measure μ on $\mathbb{R}$ such that $\mathbb{P}(\omega_x \in A) = \mu(A)$ for all $x \in \mathbb{Z}^d$ and all Borel sets $A \subset \mathbb{R}$. The family $(\omega_x)_{x \in \mathbb{Z}^d}$ is **independent** if

$$\mathbb{P}(\omega_{x_1} \in A_1, \dots, \omega_{x_l} \in A_l) = \prod_{j=1}^l \mathbb{P}(\omega_{x_j} \in A_j) = \prod_{j=1}^l \mu(A_j)$$

for every finite subset of distinct indices $\{x_1, \dots, x_l\} \subset \mathbb{Z}^d$ and arbitrary Borel sets $A_1, \dots, A_l \subset \mathbb{R}$.

In this thesis we assume that ω_x are uniformly distributed in the real closed interval $[-1, 1]$.

We consider then the probability space as the following infinite product space

$$(\Omega, \mathcal{A}, \mathbb{P}) = \bigotimes_{x \in \mathbb{Z}^d} ([-1, 1], \mathcal{B}([-1, 1]), \mu),$$

where $\mathcal{B}([-1, 1])$ is the Borel σ -algebra and $d\mu(\omega_x) = \rho(\omega_x)d\omega_x$ where the continuous density is given by $\rho = \frac{1}{2}\chi_{[-1, 1]}$.

Remark 1.1.7. It is possible to generalise the results to all absolutely continuous distributions where ρ has compact support.

As a diagonal operator, the spectrum of V_ω consists of the random entries ω_x , each corresponding to an eigenvector given by the Dirac delta

$$\delta_x \in \ell^2(\mathbb{Z}^d).$$

As for the discrete laplacian, the random term is summable in $\ell^2(\mathbb{Z}^d)$, bounded and self-adjoint. Self-adjointness of the maximal multiplication operator by a real-valued function is a fact. For definition V_ω is almost surely bounded in the closed interval $[-1, 1] \subset \mathbb{R}$, and since $\|V_\omega u\|_{\ell^2}^2 \leq \|u\|_{\ell^2}^2$ it is also summable in $\ell^2(\mathbb{Z}^d)$.

The Disorder Parameter

The disorder parameter is represented by $\lambda > 0$. The value of λ is a great indicator for localization, which makes it one of the central objects of interest of this thesis. It allows us to adjust the strength of the random disorder with respect to the deterministic part.

Let's consider the two extreme cases:

- For $\lambda = 0$ the Hamiltonian loses the random term and only considers the Laplacian. Without disorder, the material behaves like a perfect crystal without localization. The state functions are continuous plane waves and the spectrum is also continuous between $[0, 4d]$.
- For $\lambda \gg 1$ the kinetic hopping term becomes negligible compared to the random potential, so that $H_\omega \approx \lambda V_\omega$. In this scenario electrons are unable to hop between adjacent sites, and the material behaves as an insulator. In this limit, the system exhibits trivial localization: the eigenfunctions are localized Dirac deltas δ_x , and the almost sure spectrum is confined to the compact interval $[-\lambda, \lambda]$

1.2 The spectrum

The next goal is to determine the spectrum of $H_{\omega, \alpha}$. Because the random potential entries $(\omega_x)_{x \in \mathbb{Z}^d}$ are independent and identically distributed, the random Hamiltonian $H_{\omega, \alpha} = (-\Delta)^\alpha + \lambda V_\omega$ forms an ergodic family of bounded self-adjoint operators. A standard consequence of ergodic operator theory is that the spectrum of $H_{\omega, \alpha}$ is almost surely deterministic: there exists a closed, non-random set $\Sigma \subset \mathbb{R}$ such that

$$\sigma(H_{\omega, \alpha}) = \Sigma \quad \text{for } \mathbb{P}\text{-almost all } \omega \in \Omega. \quad (1.2.1)$$

More specifically, as proven in [Sto11, Theorem 2], this almost sure spectrum can be determined explicitly as the Minkowski sum of the kinetic energy spectrum and the support of the random potential distribution:

$$\Sigma = \sigma((-\Delta)^\alpha) + \lambda \operatorname{supp}(\mu). \quad (1.2.2)$$

Recall from Section 1.1 that the spectrum of the discrete Laplacian is the band $\sigma((-\Delta)^\alpha) = [0, (4d)^\alpha]$. Since the random potential V_ω has i.i.d. entries distributed uniformly on $[-1, 1]$, the almost sure spectrum of λV_ω is the interval $[-\lambda, \lambda]$. Consequently, the almost sure spectrum of H_ω is given by the compact interval:

$$\sigma(H_{\omega, \alpha}) = [0, (4d)^\alpha] + [-\lambda, \lambda] = [-\lambda, (4d)^\alpha + \lambda]. \quad (1.2.3)$$

This confirms that the spectrum is a strictly bounded subset of $\mathbb{R}$.

Preliminaries on Spectral Measures

Before characterizing the physical meaning of this spectrum, we briefly recall the relevant spectral theory. Let μ be a Borel measure on $\mathbb{R}$. Consider the set of **pure points** of μ , defined as $P = \{x \in \mathbb{R} \mid \mu(\{x\}) \neq 0\}$. We define the pure point measure as

$$\mu_{\text{pp}}(X) = \mu(P \cap X)$$

for any Borel set $X \subset \mathbb{R}$. By definition, there remains a continuous measure μ_c such that $\mu_c(\{x\}) = 0$ for all $x \in \mathbb{R}$, satisfying $\mu = \mu_{\text{pp}} + \mu_c$. Because $\mu_{\text{pp}}(P^c) = 0$ and $\mu_c(P) = 0$, these two measures are concentrated on disjoint sets and are therefore mutually singular.

This decomposition naturally extends to spectral measures:

Definition 1.2.1. Let $A : \mathcal{H} \rightarrow \mathcal{H}$ be a bounded, self-adjoint operator. For any vector $\phi \in \mathcal{H}$ and any continuous function $f \in C(\sigma(A))$, there exists a unique Borel measure μ_ϕ on the compact set $\sigma(A)$ satisfying

$$\langle \phi, f(A)\phi \rangle = \int_{\sigma(A)} f(\beta) d\mu_\phi(\beta).$$

The measure μ_ϕ is called the *spectral measure associated with the vector ϕ* .

By the spectral theorem for self-adjoint operators (see [Rud87] and [RS80]), the Hilbert space (such as $\ell^2(\mathbb{Z}^d)$) decomposes uniquely into an orthogonal

direct sum of invariant subspaces:

$$\mathcal{H} = \mathcal{H}_{\text{pp}} \oplus \mathcal{H}_{\text{c}}, \quad (1.2.4)$$

where $\mathcal{H}_{\text{pp}} = \{\phi \in \mathcal{H} \mid \mu_\phi \text{ is pure point}\}$ and $\mathcal{H}_{\text{c}} = \{\phi \in \mathcal{H} \mid \mu_\phi \text{ is continuous}\}$. For bounded self-adjoint operators, these subspaces define two (not necessarily disjoint) subsets of the spectrum:

$$\sigma(H_\omega) = \sigma_{\text{pp}}(H_\omega) \cup \sigma_{\text{c}}(H_\omega), \quad (1.2.5)$$

where $\sigma_{\text{pp}}(H_\omega) = \overline{\{\lambda \in \mathbb{R} \mid \lambda \text{ is an eigenvalue of } H_\omega\}}$.

Remark 1.2.2. In measure theory, the continuous measure μ_{c} further decomposes uniquely into $\mu_{\text{ac}} + \mu_{\text{sc}}$, where μ_{ac} is absolutely continuous with respect to Lebesgue measure, and μ_{sc} is singular continuous. Following this principle, the continuous subspace can also be decomposed as $\mathcal{H}_{\text{c}} = \mathcal{H}_{\text{ac}} \oplus \mathcal{H}_{\text{sc}}$.

Physical Interpretation and the RAGE Theorem

From a physical perspective, states in $\mathcal{H}_{\text{pp}}$ correspond to bound states that remain confined to finite regions for all time, while states in the continuous subspace $\mathcal{H}_{\text{c}}$ represent scattering states associated with transport and wave-packet spreading. This heuristic meaning is made mathematically rigorous by the RAGE theorem (named after Ruelle, Amrein, Georgescu, and Enss):

Theorem 1.2.3 (RAGE Theorem). *Let A be a self-adjoint Schrödinger operator on $\ell^2(\mathbb{Z}^d)$. Then:*

- (i) $\phi \in \mathcal{H}_{\text{pp}}$ if and only if

$$\lim_{R \rightarrow \infty} \sup_{t \in \mathbb{R}} \|\chi_{\{|x| > R\}} e^{-itA} \phi\| = 0. \quad (1.2.6)$$

- (ii) $\psi \in \mathcal{H}_{\text{c}}$ if and only if for all $R > 0$,

$$\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T \|\chi_{\{|x| \leq R\}} e^{-itA} \psi\|^2 dt = 0. \quad (1.2.7)$$

Here, $\chi_{\{|x| \leq R\}}$ and $\chi_{\{|x| > R\}}$ denote the projection operators corresponding to multiplication by the characteristic functions of the ball $\{x \in \mathbb{Z}^d \mid |x| \leq R\}$ and its complement, respectively.

In terms of this classification, Anderson localization corresponds exactly to the absence of continuous spectrum, ensuring that $\sigma(H_{\omega,\alpha}) = \sigma_{\text{pp}}(H_{\omega,\alpha})$ almost surely.

1.3 Main results

There are many studies on the Anderson localization model by both physicists and mathematicians. Various discoveries have been made over the years, yet deep challenges remain.

The main result of this thesis states that if $s \in (0, 1)$ and $\lambda > \lambda_0(s)$,

$$\mathbb{E} \left(|(H_{\omega,\alpha} - z)^{-1}(x, y)|^s \right) \begin{cases} \leq C_1 e^{-\mu|x-y|} & \text{if } \alpha = 1 \\ \leq C_2 \frac{1}{|x-y|^{d+a}} & \text{if } \alpha \in (0, 1) \end{cases} \quad (1.3.1)$$

uniformly for $z \in \mathbb{C} \setminus \mathbb{R}$ and all $x, y \in \mathbb{Z}^d$, with $C_{1,2} < \infty$ and $\mu, a > 0$. The value of $\lambda_0(s)$ changes based on the considered case. In the last decades the values of $\lambda_0(s)$ have been improved using self-avoiding walks (see [Sch15] and [DERM24]).

The resolvent is well-defined only when $z \notin \sigma(H_{\omega,\alpha})$. If z gets closer to the spectrum, the resolvent tends to infinite. The *fractional moments method* was revolutionary because averaging a fractional power of the resolvent avoids these divergences, allowing the bound to hold uniformly for all $z \in \mathbb{C} \setminus \mathbb{R}$.

Originally, the absence of conductivity was proven using squared powers instead of fractional ones. This creates divergence problems near zero and requires complicated analytical tools like the MSA method.

The intuition behind the fractional method is simple and can be understood from the following integral, which we will encounter in upcoming proofs (for example, see Lemma 3.1.4):

$$\int_{\Sigma} \frac{1}{|x|^s} dx \begin{cases} < \infty & \text{for } s \in (0, 1), \\ = \infty & \text{for } s \geq 1 \text{ (diverges at } x = 0), \end{cases}$$

where $\Sigma \subset \mathbb{R}$ is a closed interval containing the origin.

Very important results concern the value of λ . Based on the dimension of the grid, we know the following.

- $d = 1$.

For the one-dimensional grid, it has been proven, that localization holds for all $\lambda > 0$ [GMP77].

- $d = 2$.

For $d = 2$ it has been numerically established that localization persists at arbitrarily small disorder.

- $d \geq 3$.

For $d \geq 3$, to find the exact value of λ , until which the material is still a conductor, is an open problem. The best that we can prove is given in this thesis and only states, that localization shows from a specific λ_0 on (see definition in Theorem ??).

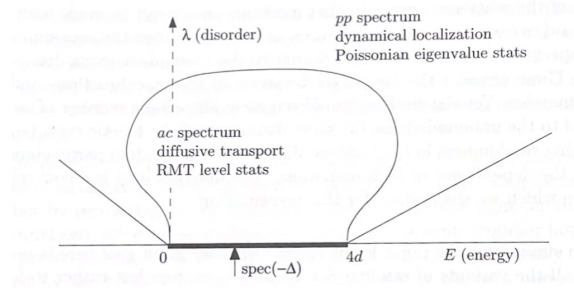


Figure 1.2: Picture from [AW15b]

Chapter 2

Localization properties

We have understood what localization means from a heuristic point of view. In the previous chapter, it was presented as a spectral property of the Hamiltonian. In this section, we introduce dynamical properties for localization and discuss the implications between them. The definitions follow those given by A. Klein in [Kle07].

2.1 Types of localization

Definition 2.1.1 (Spectral localization). We say that $H_{\omega,\alpha}$ exhibits **spectral localization** if it has pure point spectrum almost surely. That is,

$$\sigma(H_{\omega,\alpha}) = \sigma_{\text{pp}}(H_{\omega,\alpha}) \quad \text{and} \quad \sigma_{\text{c}}(H_{\omega,\alpha}) = \emptyset \quad (2.1.1)$$

for almost every realization of $\omega \in \Omega$.

Definition 2.1.2 (Dynamical localization). We say that $H_{\omega,\alpha}$ exhibits dynamical localization if for all $p > 0$ and $\phi \in \ell^2(\mathbb{Z}^d)$ of finite support,

$$\sup_{t \in \mathbb{R}} \| |X|^p e^{-itH_{\omega,\alpha}} \phi \| < \infty \quad (2.1.2)$$

for almost every realization of $\omega \in \Omega$, where the position operator $|X|$ is defined by $(|X|\phi)(n) = |n|\phi(n)$.

Remark 2.1.3. Furthermore, we can prove the following Proposition.

Proposition 2.1.4. *Let $H_{\omega,\alpha}$ be a bounded random Schrödinger operator. If (2.1.2) holds, then*

$$\sum_{x \in \mathbb{Z}^d} |x|^{2p} |e^{-itH_{\omega,\alpha}}(0, x)|^2 < \infty$$

Proof. Assume that $\psi(k) = 0$ for $|k| > R$ (i.e., ψ has finite support). Then

$$\begin{aligned} \| |X|^p e^{-itH_{\omega,\alpha}} \psi \|^2 &= \sum_x |\langle \delta_x, |X|^p e^{-itH_{\omega,\alpha}} \psi \rangle|^2 \\ &= \sum_x |x|^{2p} \left| \sum_{|k| \leq R} e^{-itH_{\omega,\alpha}}(k, x) \psi(k) \right|^2 \end{aligned}$$

Assume that the particle is initially localized in the origin (*recall: we work under translation invariance*). Then we obtain

$$= \sum_x |x|^{2p} \left| \sum_{|k| \leq R} e^{-itH_{\omega,\alpha}}(k, x) \delta_0 \right|^2 = \sum_x |x|^{2p} |e^{-itH_{\omega,\alpha}}(0, x)|^2$$

□

Definition 2.1.5 (Exponential localization). We say that $H_{\omega,\alpha}$ exhibits exponential localization if

$$\mathbb{E} \left(\sup_{t \in \mathbb{R}} |\langle \delta_0, e^{-itH_{\omega,\alpha}} \delta_x \rangle| \right) \leq C e^{-\mu|x|} \quad (2.1.3)$$

for all $x \in \mathbb{Z}^d$.

Remark 2.1.6. Dynamical localization implies that all spatial moments of the distance are bounded uniformly in time. (Recall that because we are working on an infinite grid with translation invariance, we can consider any arbitrary starting site just as easily as the origin 0).

From (2.1.2), we can already deduce that the probability of transitioning from 0 to x goes to zero faster than any polynomial, but (2.1.3) makes this explicit with a direct exponential bound. In fact, the two definitions are strongly related, and we can prove the following implication.

Proposition 2.1.7. *Exponential localization implies that all moments of the position operator are bounded in time for almost every realization of V_{ω} , i.e., it implies dynamical localization.*

Proof. To see how (2.1.2) follows from (2.1.3), we start from the result of

Proposition 2.1.4 and the fact that $|e^{-itH_{\omega,\alpha}}(0, x)| \leq 1$. It follows

$$\begin{aligned}
\mathbb{E} \left(\sup_{t \in \mathbb{R}} \| |X|^p e^{-itH_{\omega,\alpha}} \psi \|^2 \right) &\leq \sum_x |x|^{2p} \mathbb{E} \left(\sup_{t \in \mathbb{R}} |e^{-itH_{\omega,\alpha}}(0, x)|^2 \right) \quad (2.1.4) \\
&\leq \sum_x |x|^{2p} \mathbb{E} \left(\sup_{t \in \mathbb{R}} |e^{-itH_{\omega,\alpha}}(0, x)| \right) \\
&\leq C \sum_x |x|^{2p} e^{-\mu|x|} \\
&< \infty.
\end{aligned}$$

□

Remark 2.1.8. Dynamical localization holds *almost surely* rather than for every single realization because our bounds control only the *expected* value over the disorder.

Lemma 2.1.9. *Let $(X, \mathcal{A}, \mu)$ be a measure space and $f : X \rightarrow [-\infty, \infty]$ measurable. If $\int_X |f| d\mu < \infty$, then $|f| < \infty$ μ -almost everywhere, i.e. $\mu(\{x \in X : |f(x)| = \infty\}) = 0$.*

Proof. We proceed by contradiction. Let $A = \{x \in X : |f(x)| = \infty\}$ and assume $\mu(A) > 0$. Because $|f|$ is a non-negative function, the integral over the entire space X must be greater than or equal to the integral over A . Therefore,

$$\begin{aligned}
\int_X |f(x)| d\mu(x) &\geq \int_A |f(x)| d\mu(x) \\
&= \int_A \infty d\mu(x) \\
&= \infty \cdot \mu(A).
\end{aligned}$$

Since we assumed $\mu(A) > 0$, the product $\infty \cdot \mu(A)$ is exactly ∞ , which directly contradicts our initial hypothesis that $\int_X |f(x)| d\mu(x) < \infty$. Therefore $\mu(A) = 0$, which precisely means that $|f(x)| < \infty$ almost everywhere. □

Chapter 3

Fractional moments

In this section, we introduce the key lemmas and theorems that help us prove localization. The goal of this section is to demonstrate the power of the fractional moments method (FMM). First, we define the Green function, and then we prove the first fractional bound: the *a priori bound*. This will provide a good example of how we use simple but important analytic tools, such as the *resolvent identity*. Finally, we will prove that localization holds on some conditions on the disorder parameter, using the bound on the fractional power of the Green function. Most of the proof are taken from [AW15b] and [DERM24].

3.1 The Green function

The *resolvent operator* is defined as:

$$G_{z,\alpha} := (H_{\omega,\alpha} - z)^{-1} \tag{3.1.1}$$

and is a well-defined bounded operator for all $z \in \mathbb{C} \setminus \mathbb{R}$.

The following three lemmas are the most important tools in the main proofs of this thesis.

Lemma 3.1.1 (Resolvent identity). *For a linear operator $A : \mathcal{D}(A) \rightarrow \mathcal{H}$ and $z_1, z_2 \in \varrho(A)$,*

$$(A - z_1)^{-1} - (A - z_2)^{-1} = (A - z_1)^{-1}(z_1 - z_2)(A - z_2)^{-1}.$$

Moreover, if $B : \mathcal{D}(B) \rightarrow \mathcal{H}$ shares a common domain with A such that

$\mathcal{D}(A) = \mathcal{D}(B)$, then

$$(A - z)^{-1} - (B - z)^{-1} = (A - z)^{-1}(B - A)(B - z)^{-1}$$

for any $z \in \varrho(A) \cap \varrho(B)$.

The resolvent identity follows directly from simple algebraic manipulation of the inverses.

Lemma 3.1.2. For a self-adjoint operator A ,

$$\|(A - z)^{-1}\| \leq \frac{1}{\text{Im}(z)} \quad (3.1.2)$$

for all $z \in \mathbb{C} \setminus \mathbb{R}$.

Proof. For the definition of the operator norm, we know that

$$\|(A - z)^{-1}\| := \sup_{\mu \in \sigma((A - z)^{-1})} |\mu|$$

Let's consider a $\lambda \in \mathbb{C}$ and write $\mu = \frac{1}{\lambda - z}$. This means:

$$\begin{aligned} \frac{1}{\lambda - z} \in \sigma((A - z)^{-1}) &\iff (A - z)^{-1} - (\lambda - z)^{-1} \text{ not invertible} \\ &\iff (A - z)^{-1}(A - \lambda)(\lambda - z)^{-1} \text{ not invertible} \\ &\iff (A - \lambda) \text{ not invertible} \\ &\iff \lambda \in \sigma(A) \end{aligned}$$

Where the resolvent identity has been used in second implication.

Finally we obtain:

$$\|(A - z)^{-1}\| = \sup_{\lambda \in \sigma(A)} \frac{1}{|\lambda - z|} = \frac{1}{\inf_{\lambda \in \sigma(A)} |\lambda - z|} = \frac{1}{\text{dist}(\sigma(A), z)}$$

For a bounded and self-adjoint operator, we know that the spectrum is real-valued. For $z \in \mathbb{C} \setminus \mathbb{R}$ the minimum distance between z and $\sigma(A)$ is the imaginary component of z :

$$\|(A - z)^{-1}\| \leq \frac{1}{|\text{Im}(z)|} \quad (3.1.3)$$

□

From the resolvent identity we can obtain a new representation of the

Green function, which will help us develop the following proofs.

Proposition 3.1.3. *Let $H^{(\nu)} = H_{\omega,\alpha} + \nu\chi_{\{x\}}$ be a perturbed Hamiltonian in the site $x \in \mathbb{Z}^d$. Then*

$$G_z^{(\nu)}(x, y) = \frac{1}{\nu + G_{z,\alpha}(x, x)^{-1}} \cdot \frac{G_{z,\alpha}(x, y)}{G_{z,\alpha}(x, x)}. \quad (3.1.4)$$

Proof. We consider the perturbed Hamiltonian

$$H^{(\nu)} = H_{\omega,\alpha} + \nu\chi_{\{x\}}.$$

Its Green's function will be denoted by $G_z^{(\nu)}$. Applying the resolvent identity, we find

$$(H_{\omega,\alpha} - z)^{-1} = (H^{(\nu)} - z)^{-1} + \nu(H_{\omega,\alpha} - z)^{-1}\chi_{\{x\}}(H^{(\nu)} - z)^{-1}.$$

Taking the matrix elements, this yields

$$G_{z,\alpha}(x, y) = G_z^{(\nu)}(x, y) + \nu G_{z,\alpha}(x, y)G_z^{(\nu)}(x, x). \quad (3.1.5)$$

Solving for $G_z^{(\nu)}(x, y)$, we obtain

$$\begin{aligned} G_z^{(\nu)}(x, y) &= \frac{G_{z,\alpha}(x, y)}{1 + \nu G_{z,\alpha}(x, x)} \\ &= \frac{1}{\nu + G_{z,\alpha}(x, x)^{-1}} \cdot \frac{G_{z,\alpha}(x, y)}{G_{z,\alpha}(x, x)}. \end{aligned} \quad (3.1.6)$$

□

The *a priori bound* is a central estimate for the main proof in Section 4. It provides a preliminary bound for the fractional moments of the Green function, and it holds for all values of α .

Lemma 3.1.4 (A priori bound). *Let $s \in (0, 1)$ and $\alpha \in (0, 1]$. Then it holds*

$$\mathbb{E}_x(|G_{z,\alpha}(x, x)|^s) \leq \frac{1}{(1-s)\lambda^s} \quad (3.1.7)$$

for all $x \in \mathbb{Z}^d$, $z \in \mathbb{C} \setminus \mathbb{R}$, and $\lambda > 0$.

Proof of the a priori bound. For a fixed $x \in \mathbb{Z}^d$, we write $\omega = (\hat{\omega}, \omega_x)$, where $\hat{\omega} = (\omega_n)_{n \in \mathbb{Z}^d \setminus \{x\}}$. We can separate the dependence of $H_{\omega,\alpha}$ on ω_x and $\hat{\omega}$ as

$$H_{\omega,\alpha} = (-\Delta)^\alpha + \lambda V_{\hat{\omega}} + \lambda \omega_x \chi_{\{x\}} = H_{\hat{\omega},\alpha} + \lambda \omega_x \chi_{\{x\}}.$$

From Proposition 3.1.3, for $\nu = \lambda\omega_x$ in the sites (x, x) , it follows

$$G_{z,\alpha}(x, x) = \frac{1}{a + \lambda\omega_x} \quad \text{with } a = \frac{1}{\hat{G}_{z,\alpha}(x, x)}. \quad (3.1.8)$$

where $\hat{G}_{z,\alpha}$ denotes the Green function of $H_{\hat{\omega},\alpha}$.

Note that this expression is well-defined because $\hat{G}_{z,\alpha}(x, x)$ cannot be zero.

Since

$$\text{Im}\hat{G}_z(x, x) = \frac{\text{Im}(z)}{(H_{\hat{\omega},\alpha} - \text{Re}(z))^2 + \text{Im}(z)^2}(x, x)$$

always shares the same sign as $\text{Im}(z)$, whenever $\text{Im}(z) > 0$, we strictly have $\text{Im}\hat{G}_{z,\alpha}(x, x) > 0$.

The crucial fact is that the complex number a does not depend on ω_x . Thus, defining the partial expectation over the single site x as $\mathbb{E}_x(\dots) := \int \dots \rho(\omega_x) d\omega_x$, and using the uniform density $\rho(\omega_x) = \frac{1}{2}\chi_{[-1,1]}$, we find

$$\mathbb{E}_x(|G_{z,\alpha}(x, x)|^s) = \frac{1}{2\lambda^s} \int_{-1}^1 \frac{d\omega_x}{\left|\frac{a}{\lambda} + \omega_x\right|^s} \leq \frac{1}{2\lambda^s} \int_{-1}^1 \frac{d\omega_x}{|\omega_x|^s} = \frac{1}{(1-s)\lambda^s}. \quad (3.1.9)$$

The resulting constant $\frac{1}{1-s}$ is independent of λ and a , and therefore independent of the remaining random variables $\hat{\omega}$, the energy z , and the site x . $\square$

3.2 Localization at large disorder

The primary goal of this thesis is to use self-avoiding walks, combined with FMM, in order to prove the following theorems.

Theorem 3.2.1 (Schenker [Sch15]). *Let $\alpha = 1$ and $s \in (0, 1)$. Then for all $\lambda > \lambda_0(s)$,*

$$\mathbb{E}(|G_{z,\alpha}(x, y)|^s) \leq \frac{C}{(1-s)\lambda^s} e^{-\mu|x-y|} \quad \forall x, y \in \mathbb{Z}^d, x \neq y, \quad (3.2.1)$$

uniformly in $z \in \mathbb{C} \setminus \mathbb{R}$ with $C < \infty$ and $\mu > 0$.

Theorem 3.2.2 (Disertori, Escobar, Rojas-Molina [DERM24]). *Let $\alpha \in (0, 1)$ and $s \in (\frac{d}{d+2\alpha}, 1)$. Then for all $\lambda > \lambda_0(s)$*

$$\mathbb{E}[|G_{z,\alpha}(x, y)|^s] \leq \frac{K}{(1-s)\lambda^s} \frac{1}{|x-y|^{s(d+2\alpha)}} \quad \forall x, y \in \mathbb{Z}^d, x \neq y, \quad (3.2.2)$$

uniformly in $z \in \mathbb{C} \setminus \mathbb{R}$ with $K < \infty$.

Remark 3.2.3. In both theorems, we encounter the condition on the disorder parameter $\lambda > \lambda_0(s)$. For both scenarios, $\lambda_0(s) := \left(\frac{1}{(1-s)R_{\chi_D}} \right)$. In particular, for $\alpha = 1$, it has been proven by Schenker in [Sch15], that $\lambda_0(s)$ is the unique solution of $\lambda_0 = \mu_d e \ln(\lambda_0)$, where $R_{\chi_D} = 1/\mu_d$ is a definition from the self-avoiding walks representations (see Section 4).

3.2.1 From fractional moments to localization

In this subsection we aim to prove localization, namely spectral localization for $\alpha \in (0, 1]$ (Theorem 3.2.5), and exponential localization for $\alpha = 1$ (Theorem 3.2.9). In particular, we will also see that for $\alpha \in (0, 1)$ a partial form of dynamical localization holds (Theorem 3.2.10).

Spectral localization

We obtain spectral localization for all $\alpha \in (0, 1]$ through the Simon-Wolff criterion.

Proposition 3.2.4 (Simon-Wolff criterion). *Let $H_{\omega, \alpha}$ be a random operator in $l^2(\mathbb{Z}^d)$ with a random potential whose conditional-single site distribution is absolutely continuous. Then if for Lebesgue-almost every $E \in \mathbb{R}$ and $\mathbb{P}$ -almost all ω :*

$$\lim_{\eta \downarrow 0} \sum_{y \in \mathbb{Z}^d} |G_{E+i\eta, \alpha}(x, y)|^2 < \infty \quad (3.2.3)$$

$H_{\omega, \alpha}$ has only pure point spectrum.

Proof can be found in [AW15b, Chapter 5]

Theorem 3.2.5. *Let Theorem 3.2.1 and 3.2.2 and hold. Then, the spectrum of $H_{\omega, \alpha}$ consists only of pure point spectrum for a.e. realisation of V_ω .*

Proof from [DERM24]. From the Theorems 3.2.1 and 3.2.2 we notice that for all $\lambda > \lambda_0$:

$$\sum_{y \in \mathbb{Z}^d} \mathbb{E}[|G_{E+i\eta, \alpha}(x, y)|^s] = \frac{C}{(1-s)\lambda^s} \sum_{y \in \mathbb{Z}^d} e^{-\mu|x-y|} < \infty \quad \text{for all } \mu > 0 \quad (3.2.4)$$

and

$$\begin{aligned} \sum_{y \in \mathbb{Z}^d} \mathbb{E}[|G_{E+i\eta, \alpha}(x, y)|^s] &= \frac{K}{(1-s)\lambda^s} \sum_{y \in \mathbb{Z}^d} \frac{1}{|x-y|^{s(d+2\alpha)}} \\ &< \infty \quad \text{for all } s \in \left(\frac{d}{d+2\alpha}, 1\right) \end{aligned} \quad (3.2.5)$$

uniformly for $z \in \mathbb{C} \setminus \mathbb{R}$ and $x \in \mathbb{Z}^d$ fixed.

From Proposition ?? we notice also that $\sum_{y \in \mathbb{Z}^d} \mathbb{E}[|G_{E+i\eta, \alpha}(x, y)|^2]$ always exists but might be infinite for $\eta \rightarrow 0$. We argue:

$$\begin{aligned} \mathbb{E} \left[\left(\lim_{\eta \downarrow 0} \sum_{y \in \mathbb{Z}^d} |G_{E+i\eta, \alpha}(x, y)|^2 \right)^{\frac{s}{2}} \right] &= \mathbb{E} \left[\lim_{\eta \downarrow 0} \left(\sum_{y \in \mathbb{Z}^d} |G_{E+i\eta, \alpha}(x, y)|^2 \right)^{\frac{s}{2}} \right] \\ &\leq \mathbb{E} \left[\lim_{\eta \downarrow 0} \sum_{y \in \mathbb{Z}^d} |G_{E+i\eta, \alpha}(x, y)|^s \right] \\ &\leq \liminf_{\eta \downarrow 0} \sum_{y \in \mathbb{Z}^d} \mathbb{E}[|G_{E+i\eta, \alpha}(x, y)|^s] < \infty, \end{aligned}$$

where in the first two steps we used that the function $x \mapsto x^{\frac{s}{2}}$ is monotone and $|\sum_n a_n|^s \leq \sum_n |a_n|^s$ for all $a_n \geq 0$ and $0 < s < 1$. The last two inequalities follow by Fatou and (3.2.4) and (3.2.5). $\square$

Dynamical localization

In order to prove the next theorems, we need the following definition.

Definition 3.2.6. Let Σ be the almost sure spectrum of $H_{\omega, \alpha}$. Then we call

$$Q(x, y, \Sigma) := \sup_{\substack{g \in C_c(\Sigma) \\ |g| \leq 1}} |g(H_{\omega, \alpha})(x, y)| \quad (3.2.6)$$

the *eigenfunction correlator*, for all $x, y \in \mathbb{Z}^d$.

Remark 3.2.7. Note that

$$\sup_{t \in \mathbb{R}} |e^{-itH_{\omega, \alpha}}(0, x)|^2 \leq \sup_{\substack{g \in C_c(\Sigma) \\ |g| \leq 1}} |g(H_{\omega, \alpha})(0, x)| =: Q(x, 0, \Sigma)$$

The following Proposition, is the main step for the proofs of Theorem 3.2.9 and Theorem 3.2.10.

Proposition 3.2.8. *Let Σ be the almost sure spectrum of $H_{\omega,\alpha}$. Then,*

$$\mathbb{E}[Q(x, y; \Sigma)] \leq C_s \lim_{|\eta| \downarrow 0} \int_{\Sigma} \mathbb{E}[|G(x, y; E + i\eta)|^s] dE \quad (3.2.7)$$

with a finite constant $C_s < \infty$

Theorem 3.2.9. *Let Σ be the almost sure spectrum of $H_{\omega,\alpha}$. Then for $\alpha = 1$ we obtain exponential (and consequently dynamical) localization.*

Proof. We apply Theorem 3.2.1 and Proposition 3.2.8

$$\mathbb{E}[Q(x, 0, \Sigma)] \leq C_s \frac{C}{(1-s)\lambda^s} \lim_{|\eta| \downarrow 0} \int_{\Sigma} \mathbb{E}[e^{-\mu|x-y|}] dE \leq C_s |\Sigma| \frac{C}{(1-s)\lambda^s} e^{-\mu|x-y|} \quad (3.2.8)$$

as we wanted. $\square$

Theorem 3.2.10. *Let Σ be the almost sure spectrum of $H_{\omega,\alpha}$. Then for $\alpha \in (0, 1)$ and $0 < \beta < 2\alpha_s$, we have for a.e. $\omega \in \Omega$*

$$\sup_{t \in \mathbb{R}} \sum_{x \neq 0} |x|^\beta |e^{-itH_{\alpha,\omega}}(0, x)|^2 < \infty. \quad (3.2.9)$$

In particular, if $\alpha > \frac{1}{2}$, the first moment of the wave function initially localized at the origin and evolving under the action of $H_{\alpha,\omega}$ is finite.

Proof. We apply Theorem 3.2.2 and Proposition 3.2.8, and obtain

$$\mathbb{E}[Q(x, 0, \Sigma)] \leq \frac{C_s K}{(1-s)\lambda^s} \sum_{x \neq 0} |x|^\beta \frac{1}{|x|^{d+2\alpha_s}} < \infty,$$

which converges for all $\beta < s(d+2\alpha) - d$. Since $0 \leq Q(x, 0, \Sigma) \leq 1$, we can conclude that for almost every $\omega \in \Omega$:

$$\sup_{t \in \mathbb{R}} \sum_{x \neq 0} |x|^\beta |e^{-itH_{\alpha,\omega}}(0, x)|^2 \leq \sum_{x \neq 0} |x|^\beta Q_\omega(x, 0)^2 \leq \sum_{x \neq 0} |x|^\beta Q_\omega(x, 0) < \infty,$$

which yields the required dynamical bounds. $\square$

Proof of Proposition 3.2.8. As we mentioned in Definition 3.2.6, for bounded sets $\Sigma \subset \mathbb{R}$

$$Q(x, y, \Sigma) := \sup_{\substack{g \in C_c(\Sigma) \\ |g| \leq 1}} |g(H_{\omega,\alpha})(x, y)|$$

. For $g \in C_c(\Sigma)$ and $\xi \in \mathbb{R}$

$$g(\xi) = \lim_{\eta \downarrow 0} \frac{\eta}{\pi} \int \frac{g(E)}{(\xi - E)^2 + \eta^2} dE.$$

which for $|g| \leq 1$ means

$$Q(x, y, \Sigma) \leq \lim_{\eta \downarrow 0} \frac{1}{\pi} \int_{\Sigma} |\operatorname{Im}(G_{E+i\eta, \alpha})(x, y)| dE \quad (3.2.10)$$

Averaging a positive function (see proof of *a priori bound* 3.1.4), and for this reason applying Fatou's and Fubini's theorems gives

$$\mathbb{F}[Q(x, y; I)] \leq \liminf_{\eta \downarrow 0} \frac{1}{\pi} \int_I \mathbb{E}[|\operatorname{Im}(G_{E+i\eta, \alpha})(x, y)|] dE. \quad (3.2.11)$$

By the Cauchy-Schwarz inequality

$$|\operatorname{Im}(G_{E+i\eta, \alpha})(x, y)| \leq \sqrt{|\operatorname{Im}(G_{E+i\eta, \alpha})(x, x)| |\operatorname{Im}(G_{E+i\eta, \alpha})(y, y)|}. \quad (3.2.12)$$

where $\operatorname{Im}(G_{E+i\eta, \alpha})(x, y) > 0$ when $\eta > 0$.

Splitting the integrand in (3.2.11) into powers $(1-s)$ and s and applying (3.2.12) to just the s power we get

$$\begin{aligned} & \left(\int_I \mathbb{E}[|\operatorname{Im}(G_{E+i\eta, \alpha})(x, y)|] dE \right)^2 \\ &= \left(\int_I \mathbb{E}[|\operatorname{Im}(G_{E+i\eta, \alpha})(x, y)|^{1-s} |\operatorname{Im}(G_{E+i\eta, \alpha})(x, y)|^s] dE \right)^2 \\ &\leq \left(\int_I \mathbb{E} \left[\left(|\operatorname{Im}(G_{E+i\eta, \alpha})(x, x)|^{\frac{1-s}{2}} |\operatorname{Im}(G_{E+i\eta, \alpha})(y, y)|^{\frac{1-s}{2}} \right) |\operatorname{Im}(G_{E+i\eta, \alpha})(x, y)|^s \right] dE \right)^2 \\ &\leq \int_I \mathbb{E}[|\operatorname{Im}(G_{E+i\eta, \alpha})(x, x)|^{1-s} |\operatorname{Im}(G_{E+i\eta, \alpha})(x, y)|^s] dE \\ &\quad \times \int_I \mathbb{E}[|\operatorname{Im}(G_{E+i\eta, \alpha})(y, y)|^{1-s} |\operatorname{Im}(G_{E+i\eta, \alpha})(x, y)|^s] dE, \end{aligned} \quad (3.2.13)$$

where we again applied the Cauchy-Schwarz inequality. Since we are working under translation invariance, we focus now on bounding one of the integrals, knowing that everything holds for the other too.

Using the subadditivity property $|\sum_n a_n|^s \leq \sum_n |a_n|^s$ we estimate

$$|\operatorname{Im}(G_{E+i\eta, \alpha})(x, y)|^s \leq \frac{1}{2^s} (|G_{E+i\eta, \alpha}(x, y)|^s + |G_{E-i\eta, \alpha}(x, y)|^s). \quad (3.2.14)$$

Since $|\operatorname{Im}(G_{E+i\eta,\alpha})(x, x)| = |\operatorname{Im}(G_{E-i\eta,\alpha})(x, x)|$, the results that we obtain for the first term, are the same as for the second and we can finally bound

$$I := |\operatorname{Im}(G_{E+i\eta,\alpha})(x, x)|^{1-s} |G_{E+i\eta,\alpha}(x, y)|^s. \quad (3.2.15)$$

Now, we consider Proposition 3.1.3, and notice that

$$\frac{G_z^{(\nu)}(x, y)}{G_z^{(\nu)}(x, x)} = \frac{\frac{G_{z,\alpha}(x, y)}{1+\nu G_{z,\alpha}(x, x)}}{\frac{G_{z,\alpha}(x, x)}{1+\nu G_{z,\alpha}(x, x)}} = \frac{G_{z,\alpha}(x, y)}{G_{z,\alpha}(x, x)}.$$

For $\nu = \zeta - \lambda\omega_x$, we obtain the operator $H_\zeta := H_{\omega,\alpha} + (\zeta - \lambda\omega_x)\chi_{\{x\}}$.

In particular, we recall from the *a priori bound* (Lemma 3.1.4)

$$\hat{H}_{\omega,\alpha} := H - \lambda\omega_x \chi_{\{x\}} \quad \text{und} \quad G_{z,\alpha}(x, x) = \frac{1}{\hat{G}_{z,\alpha}(x, x)^{-1} + \lambda\omega_x}.$$

This implies for (3.2.15) that

$$\begin{aligned} & |\operatorname{Im}(G_{z,\alpha}(x, x))|^{1-s} |G_{z,\alpha}(x, y)|^s \\ &= \left| \operatorname{Im} \left(\frac{1}{\hat{G}_{z,\alpha}(x, x)^{-1} + \lambda\omega_x} \right) \right|^{1-s} \left| \frac{1}{\hat{G}_{z,\alpha}(x, x)^{-1} + \lambda\omega_x} \frac{G_z^{(\zeta)}(x, y)}{G_z^{(\zeta)}(x, x)} \right|^s \\ &= \frac{|\operatorname{Im}(\hat{G}_{z,\alpha}(x, x)^{-1})|^{1-s}}{|\hat{G}_{z,\alpha}(x, x)^{-1} + \lambda\omega_x|^{2-2s}} \frac{|\zeta + \hat{G}_{z,\alpha}(x, x)^{-1}|^s}{|\hat{G}_{z,\alpha}(x, x)^{-1} + \lambda\omega_x|^s} |G_z^{(\zeta)}(x, y)|^s \\ &\leq (|\zeta|^s + |\hat{G}_{z,\alpha}(x, x)^{-1}|^s) \frac{|\operatorname{Im}(\hat{G}_{z,\alpha}(x, x)^{-1})|^{1-s}}{|\hat{G}_{z,\alpha}(x, x)^{-1} + \lambda\omega_x|^{2-s}} |G_z^{(\zeta)}(x, y)|^s \end{aligned}$$

where we used the triangle inequality in the last step.

Now we compute the conditional expectation with respect to the random variable ω_x . Let us denote the distribution of $\lambda\omega_x$ by $\rho_\lambda(v) dv$. Taking the conditional expectation given $\{\omega_y\}_{y \neq x}$, we integrate over v :

$$\begin{aligned} \mathbb{E}[I \mid \{\omega_y\}_{y \neq x}] &= \int_{\mathbb{R}} I(v) \rho_\lambda(v) dv \\ &\leq (|\zeta|^s + |\hat{G}_{z,\alpha}(x, x)^{-1}|^s) |G_z^{(\zeta)}(x, y)|^s \\ &\quad \times \int_{\mathbb{R}} \frac{|\operatorname{Im}(-\hat{G}_{z,\alpha}(x, x)^{-1})|^{1-s}}{|v - (-\hat{G}_{z,\alpha}(x, x)^{-1})|^{2-2s}} \frac{\rho_\lambda(v)}{|v - (-\hat{G}_{z,\alpha}(x, x)^{-1})|^s} dv. \end{aligned} \quad (3.2.16)$$

At this point, Lemma 3.2.11 is applied to the remaining integral with $\gamma = -\hat{G}_{z,\alpha}(x, x)^{-1}$. The integral completely absorbs the $(1-s)$ powers and is bounded by $C_s / \max\{1, |\gamma|^s\}$. The multiplicative factor $(|\zeta|^s + |\gamma|^s)$ is thus reduced to $(|\zeta|^s + 1)$ thanks to this normalization

$$\mathbb{E} [|\operatorname{Im}(G_{z,\alpha}(x, x))|^{1-s} |G_{z,\alpha}(x, y)|^s] \leq C_s (|\zeta|^s + 1) |G_z^{(\zeta)}(x, y)|^s. \quad (3.2.17)$$

Because the inequality (3.2.17) holds for any arbitrary $\zeta \in \mathbb{R}$, the resulting bound may now be averaged over ζ . For its probability distribution, we pick a density proportional to $(|\zeta|^s + 1)^{-1} \rho_\lambda(\zeta)$. Note that the normalization constant for this distribution is strictly positive. Averaging over this distribution neutralizes the polynomial prefactor $(|\zeta|^s + 1)$, and taking the expectation over the remaining variables yields

$$\mathbb{E} [|\operatorname{Im}(G_{E+i\eta,\alpha}(x, x))|^{1-s} |G_{E+i\eta,\alpha}(x, y)|^s] \leq C'_{s,\lambda} \mathbb{E} [|G_{E+i\eta,\alpha}(x, y)|^s]. \quad (3.2.18)$$

As noted earlier, we also need to bound the second factor in (3.2.13) which involves $E - i\eta$. Because $|\operatorname{Im}(G_{E-i\eta,\alpha}(y, y))| = |\operatorname{Im}(G_{E+i\eta,\alpha}(y, y))|$ for any complex energy off the real axis, the exact same upper bound rigorously applies to the second factor.

Substituting these bounds back into the two factors of our original Cauchy-Schwarz inequality (3.2.13), we obtain:

$$\left(\int_{\Sigma} \mathbb{E} [|\operatorname{Im}(G_{E+i\eta,\alpha}(x, y))|] dE \right)^2 \leq (C'_{s,\lambda})^2 \left(\int_{\Sigma} \mathbb{E} [|G_{E+i\eta,\alpha}(x, y)|^s] dE \right)^2.$$

Taking the square root of both sides and substituting this result back into (3.2.11), we can take the limit $|\eta| \downarrow 0$:

$$\mathbb{E}[Q(x, y; \Sigma)] \leq C'_{s,\lambda} \liminf_{|\eta| \downarrow 0} \int_{\Sigma} \mathbb{E} [|G_{E+i\eta,\alpha}(x, y)|^s] dE. \quad (3.2.19)$$

This concludes the proof. $\square$

Lemma 3.2.11. *Let $0 \leq \rho$ with $\sup_{v \in \mathbb{R}} (1 + |v|^s) \rho(v) < \infty$ for some $s \in (0, 1)$. Then there exists some constant $C_s(\rho) < \infty$ such that for any $\gamma \in \mathbb{C} \setminus \mathbb{R}$,*

$$\max\{1, |\gamma|^s\} \int \frac{|\operatorname{Im} \gamma|^{1-s}}{|v - \gamma|^{2-2s}} \frac{\rho(v)}{|v - \gamma|^s} dv \leq C_s(\rho). \quad (3.2.20)$$

Proof. Look [AW15b, Lemma 7.8]. $\square$

Chapter 4

SAW representation

In this Chapter, we are going to use the SAW representation method, in order to prove Theorem 3.2.1 and Theorem 3.2.2. At first, we will give the main definitions regarding the self-avoiding walks, which are all generated by the *step weight kernel* $D(x, y)$ (see Definition 1.1.3). Secondly, we develop a new representation possibility for the Green function using SAW. At the end, we prove the two Theorems.

4.1 SAW Definitions

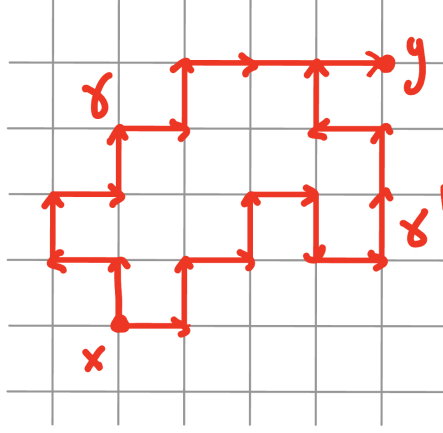
Definition 4.1.1 (Self-Avoiding Walk). A **self-avoiding walk** on the integer lattice $\mathbb{Z}^d$ of length $N \in \mathbb{N}_0$ is an ordered $(N + 1)$ -tuple of distinct lattice sites

$$\gamma = (\gamma_0, \gamma_1, \dots, \gamma_N) \in (\mathbb{Z}^d)^{N+1}$$

such that $\gamma_i \neq \gamma_j$ for all $i \neq j$.

Definition 4.1.2 (Sets of Self-Avoiding Walks). Let $\Lambda \subseteq \mathbb{Z}^d$. For any $x, y \in \Lambda$ and $N \in \mathbb{N}_0$:

- (i) $\mathcal{S}_N^{(\Lambda)}(y, x)$ denotes the set of all self-avoiding walks of length N contained entirely in Λ , starting at $\gamma_0 = x$ and terminating at $\gamma_N = y$. When $\Lambda = \mathbb{Z}^d$, we simply write $\mathcal{S}_N(y, x)$.
- (ii) $\mathcal{S}_N^{(\Lambda)}(x) := \bigcup_{y \in \Lambda} \mathcal{S}_N^{(\Lambda)}(y, x)$ denotes the set of all self-avoiding walks of length N in Λ starting at x .


 Figure 4.1: γ and γ' are self-avoiding walks from x to y

- (iii) $c_N^D := \sum_x \sum_{\gamma \in \mathcal{S}_N(0,x)} \prod_{j=0}^{N-1} D(\gamma_j, \gamma_{j+1})$ denotes the total number of self-avoiding walks of length N on $\mathbb{Z}^d$ originating from a fixed site (e.g., the origin 0), which is independent of the starting vertex by translation invariance.

Definition 4.1.3. (Correlation Function) For a parameter $\beta \geq 0$, the self-avoiding walk correlation function $C_\beta^D(y - x)$ between two vertices $x, y \in \mathbb{Z}^d$ is the weighted generating function:

$$C_\beta^D(y - x) := \sum_{N=0}^{\infty} \beta^N \sum_{\gamma \in \mathcal{S}_N(x,y)} \prod_{j=0}^{N-1} D(\gamma_j, \gamma_{j+1})$$

defined for all values of β for which the power series is absolutely convergent. By translation invariance, $C_\beta^D(y - x) = C_\beta^D(0, y - x) = C_\beta^D(x, y)$ depends only on $y - x$.

Definition 4.1.4. (Susceptibility) The susceptibility $\chi^D(\beta)$ of self-avoiding walks is the sum of the two-point correlation function over all endpoints:

$$\chi^D(\beta) := \sum_{x \in \mathbb{Z}^d} C_\beta^D(x) = \sum_{N=0}^{\infty} \beta^N c_N^D$$

The series converges uniformly and absolutely if and only if $\beta < R_{\chi^D}$.

Definition 4.1.5. (Radius of Convergence) R_{χ^D} is critical radius of convergence of the susceptibility with respect to the kernel D .

This is defined by

$$R_{\chi^D} := \frac{1}{\lim_{N \rightarrow \infty} (c_N^D)^{1/N}}.$$

This limit exists, is positive, and satisfies a strict inequality dependent on the summability of the fractional kernel.

Claim 4.1.6. Let R_{χ^D} be the radius of convergence of self-avoiding walks on $\mathbb{Z}^d$ generated by D . Then

$$R_{\chi^D} \geq \frac{1}{\sum_{z \neq 0} D(0, z)} > 0$$

Proof. Consider the generating function for the weights D . For $N = 2$, we calculate the combinations of long-range jumps

$$\sum_{x \in \mathbb{Z}^d} \sum_{\gamma \in \mathcal{S}_2(0, x)} D(0, \gamma_1) D(\gamma_1, x) \leq \left(\sum_{w \neq 0} D(0, w) \right)^2$$

Repeating this argument inductively for a general length $N \geq 1$, we obtain an upper bound based on the geometric properties of the series

$$\sum_{x \in \mathbb{Z}^d} \sum_{\gamma \in \mathcal{S}_N(0, x)} \prod_{j=0}^{N-1} D(\gamma_j, \gamma_{j+1}) \leq \left(\sum_{z \neq 0} D(0, z) \right)^N$$

Since the geometric series converges provided that the product with the parameter β is strictly less than 1, the radius of convergence satisfies the required inequality $R_{\chi^D} \geq \left(\sum_{z \neq 0} D(0, z) \right)^{-1}$. $\square$

Remark 4.1.7. For $\alpha = 1$ the radius of convergence is defined as $1/\mu_d$, where

$$\mu_d := \lim_{N \rightarrow \infty} (c_N)^{1/N}.$$

and $\mu_d < 2d$. one can prove that $\mu_d < 2d - 1$ APPENDIX

Claim 4.1.8. Let μ_d be the connective constant (or growth rate) of self-avoiding walks on $\mathbb{Z}^d$. Then:

$$\mu_d \leq 2d - 1$$

Proof. Let c_N denote the number of self-avoiding walks (SAWs) of length N starting from the origin $0 \in \mathbb{Z}^d$, and let $S_N(0, x)$ denote the set of such walks

ending at $x \in \mathbb{Z}^d$. By definition:

$$c_N = \sum_{x \in \mathbb{Z}^d} |S_N(0, x)| = \sum_{x \in \mathbb{Z}^d} \sum_{\gamma \in S_N(0, x)} 1$$

Consider the generating function (*susceptibility*) with parameter $\beta > 0$ in $\mathbb{C}$:

$$\sum_{N=0}^{\infty} |\beta|^N c_N = \sum_{N=0}^{\infty} |\beta|^N \sum_{x \in \mathbb{Z}^d} |S_N(0, x)|$$

For $N = 0$, $c_0 = 0$. For $N \geq 1$, a walk cannot immediately step back onto the previous vertex (non-reversing condition). Thus:

- The first step has at most $2d$ choices.
- Each of the subsequent $N - 1$ steps has at most $2d - 1$ choices.

This gives the upper bound:

$$c_N \leq 2d(2d - 1)^{N-1}, \quad \text{for } N \geq 1.$$

Using this upper bound, we can bound the series from above:

$$\begin{aligned} \sum_{N=0}^{\infty} |\beta|^N c_N &\leq \sum_{N=1}^{\infty} |\beta|^N (2d)(2d - 1)^{N-1} \\ &= 2d|\beta| \sum_{k=0}^{\infty} |\beta|^k (2d - 1)^k \quad (\text{setting } k = N - 1) \\ &= 2d|\beta| \sum_{k=0}^{\infty} (|\beta|(2d - 1))^k \end{aligned}$$

This geometric series converges if and only if:

$$|\beta|(2d - 1) < 1 \iff |\beta| < \frac{1}{2d - 1} =: R$$

where $R = \frac{1}{2d-1}$ is the radius of convergence of the bounding series.

By the Cauchy–Hadamard theorem, the radius of convergence R' of the original series $\sum_{N=0}^{\infty} c_N |\beta|^N$ is:

$$R' = \frac{1}{\limsup_{N \rightarrow \infty} c_N^{1/N}} = \frac{1}{\mu_d}$$

Since the original series is bounded term-by-term by a series that converges

for $|\beta| < R$, the true radius of convergence must satisfy:

$$R' \geq R \implies \frac{1}{\mu_d} \geq \frac{1}{2d-1}$$

Taking the reciprocal yields:

$$\mu_d \leq 2d - 1.$$

□

The final results regarding the decay of the fractional moments of the green function, depend on the estimated decay of the two point function. In fact we can prove the following Propositions.

Proposition 4.1.9 (Polynomial Decay of the Correlation Function). *Let $\alpha \in (0, 1)$, $s \in (\frac{d}{d+2\alpha}, 1)$ and $\beta \geq 0$ such that $\beta < R_{\chi^D}$. Then there exists a constant $K_0 < \infty$, independent of x , such that*

$$C_\beta^D(x) \leq K_0 \frac{1}{|x|^{s(d+2\alpha)}} \quad \text{for all } x \in \mathbb{Z}^d.$$

The proof of this Lemma can be found in [DERM24, Lemma 4].

Proposition 4.1.10 (Exponential Decay of the Correlation Function). *Let $\alpha = 1$, $s \in (0, 1)$ and $\beta \geq 0$ such that $\beta < 1/\mu_d$. Then for every $\varepsilon > 0$ there exists a constant $K_\varepsilon < \infty$, independent of x , such that*

$$C_\beta^D(x) \leq K_\varepsilon ((\mu_d + \varepsilon)\beta)^{|x|} \quad \text{for all } x \in \mathbb{Z}^d.$$

Proof. We know that $|\mathcal{S}_N(0, x)| = 0$ whenever $N < |x|$, and the sum defining $C_\beta^D(x)$ may be restricted to $N \geq |x|$:

$$C_\beta^D(x) = \sum_{N=|x|}^{\infty} \beta^N |\mathcal{S}_N(x, 0)| \leq \sum_{N=|x|}^{\infty} \beta^N c_N. \quad (4.1.1)$$

Fix $\varepsilon > 0$. Since $\mu_d = \lim_{N \rightarrow \infty} c_N^{1/N}$, by the definition of limit there exists $N_0 \in \mathbb{N}$ such that

$$c_N^{1/N} < \mu_d + \varepsilon \quad \text{for all } N \geq N_0,$$

that is,

$$c_N < (\mu_d + \varepsilon)^N \quad \text{for all } N \geq N_0.$$

For $N = 0, 1, \dots, N_0 - 1$, define

$$K_\varepsilon := \max \left(1, \max_{0 \leq N < N_0} \frac{c_N}{(\mu_d + \varepsilon)^N} \right),$$

which is finite, being the maximum of finitely many finite quantities. By construction it holds:

$$c_N \leq K_\varepsilon (\mu_d + \varepsilon)^N \quad \text{for all } N \geq 0.$$

Combining it with (4.1.1) we obtain

$$C_\beta^D(x) \leq K_\varepsilon \sum_{N=|x|}^{\infty} [(\mu_d + \varepsilon)\beta]^N.$$

Choosing ε small enough that $r := (\mu_d + \varepsilon)\beta < 1$ (possible since $\beta < 1/\mu_d$), the tail of the geometric series equals

$$\sum_{N=|x|}^{\infty} r^N = \frac{r^{|x|}}{1-r}.$$

Hence

$$C_\beta^D(x) \leq \frac{K_\varepsilon}{1 - (\mu_d + \varepsilon)\beta} [(\mu_d + \varepsilon)\beta]^{|x|}.$$

where $\frac{K_\varepsilon}{1 - (\mu_d + \varepsilon)\beta}$ is still finite and independent from x . $\square$

4.2 SAW representation of G_z

The proof of the theorem 3.1 is similar to the one given by Schenker in [1]. We are going to work with the Green function defined on a volume where one site is removed every stage. Therefore, for any $\Lambda \subseteq \mathbb{Z}^d$ (finite or infinite) the Green function changes:

$$G_z^\Lambda(x, y) := \begin{cases} (H_\omega^\Lambda - z), & \forall x, y \in \Lambda \\ 0, & \text{otherwise} \end{cases} \quad (4.2.1)$$

where $H_\omega^\Lambda - z \in \mathbb{C}^{\Lambda \times \Lambda}$ is the matrix obtained by restricting $H_\omega - z$ to Λ . In this thesis we will consider $\Lambda = \mathbb{Z}^d$, but we leave the notation Λ to highlight the fact that the same result holds for any volume.

To notice is also that $H_\omega^\Lambda - z$ is still invertible for all $z \in \mathbb{C} \setminus \mathbb{R}$. In this

section, for simplicity during calculations, we will write $G^\Lambda(x, y)$ instead of $G_z^\Lambda(x, y)$.

Theorem 4.2.1. *Let $x, y \in \mathbb{Z}^d$, $x \neq y$, $\Lambda \subseteq \mathbb{Z}^d$. G^Λ is defined as in (4.2.1). It holds*

$$G^\Lambda(x, y) = \sum_{n=1}^N \sum_{\gamma \in \mathcal{S}_n(x, y)} \prod_{i=1}^n G^\Lambda(x, x) \cdot G^{\Lambda_i}(\gamma_i, \gamma_i) + R_N \quad (4.2.2)$$

where $\Lambda_i = \Lambda \setminus \{x, \gamma_1, \dots, \gamma_{i-1}\}$ and

$$R_N = \sum_{x \sim y} \sum_{\gamma \in \mathcal{S}_N(0, x)} \prod_{i=1}^{N-1} G^\Lambda(x, x) \cdot G^{\Lambda_i}(\gamma_i, \gamma_i) \cdot G^{\Lambda_N}(\gamma_N, y). \quad (4.2.3)$$

In order to prove Theorem 4.2.1, we first need the following claim to hold.

Claim 4.2.2.

$$G^\Lambda(x, y) = G^\Lambda(x, x) \sum_{x_1 \sim x} G^{\Lambda_1}(x_1, y) \quad (4.2.4)$$

Proof. Using the resolvent identity for $A = (H_\omega^\Lambda - z)$ and $B = (H_\omega^{\{x\}} \oplus H_\omega^{\Lambda \setminus \{x\}} - z)$, we obtain:

$$\begin{aligned} & (H_\omega^\Lambda - z)^{-1} - (H_\omega^{\{x\}} \oplus H_\omega^{\Lambda \setminus \{x\}} - z)^{-1} \\ &= -(H_\omega^\Lambda - z)^{-1} \left(H_\omega^\Lambda - (H_\omega^{\{x\}} \oplus H_\omega^{\Lambda \setminus \{x\}}) \right) (H_\omega^{\{x\}} \oplus H_\omega^{\Lambda \setminus \{x\}} - z)^{-1} \end{aligned}$$

Moreover:

$$H_\omega^\Lambda - (H_\omega^{\{x\}} \oplus H_\omega^{\Lambda \setminus \{x\}}) = - \sum_{x_1 \sim x} \delta_{x_1} =: T_{x, x_1}$$

Now considering the matrix elements in (x, y) with $x \neq y$ and matrix multiplication:

$$\langle \delta_x, (H_\omega^\Lambda - z)^{-1} \delta_y \rangle - 0 = - \sum_{x_1, x_2} \langle \delta_x, (H_\omega^\Lambda - z)^{-1} \delta_{x_1} \rangle T_{x_1, x_2} \langle \delta_{x_2}, (H_\omega^{\{x\}} \oplus H_\omega^{\Lambda \setminus \{x\}} - z)^{-1} \delta_y \rangle$$

For $T_{x_1, x_2} \neq 0$, we need $x_1 = x$ and $x_2 \sim x_1$.

Which leaves us with:

$$G^\Lambda(x, y) = \sum_{x_1 \sim x} G^\Lambda(x, x) G^{\Lambda \setminus \{x\}}(x_1, y)$$

□

Proof of Theorem 5.8. Now we separate the case in which $x_1 = y$ (which implies $|x - y| = 1$).

$$G^\Lambda(x, y) = G^\Lambda(x, x) \left(\sum_{x_1 \sim x, x_1 \neq y} G^{\Lambda_1}(x_1, y) + \delta_{|x-y|=1} G^{\Lambda_1}(y, y) \right)$$

Iterating N times, it gives (4.2.2). □

4.3 Proof with SAW

In this last Section, we can use the SAW representation of the Green function (4.2.2), in order to prove the following Theorem.

Theorem 4.3.1. *Let $\alpha \in (0, 1]$. Then, for all the above definitions, it holds*

$$\mathbb{E}(|G_{\alpha, \omega}(x, y)|^s) \leq \frac{1}{\lambda^s(1-s)} C_\beta^D(y - x) \quad (4.3.1)$$

for all $x, y \in \mathbb{Z}^d$

Proof. Taking the fractional moment of both sides of the resolvent identity (4.2), and applying the subadditivity inequality $|\sum a_i|^s \leq \sum |a_i|^s$ (which holds for $s \in (0, 1)$), we obtain:

$$\begin{aligned} \mathbb{E} [|G_z^\Lambda(x, y)|^s] &\leq \sum_{n=1}^N \sum_{\gamma \in \mathcal{S}_n(x, y)} \left(\prod_{i=1}^n D(\gamma_{i-1}, \gamma_i) \right) \mathbb{E} \left[|G_z^\Lambda(x, x)|^s \prod_{i=1}^n |G_z^{\Lambda_i}(\gamma_i, \gamma_i)|^s \right] \\ &\quad + \sum_{w \in \mathbb{Z}^d} \sum_{\gamma \in \mathcal{S}_{N+1}(x, w)} \left(\prod_{i=1}^{N+1} D(\gamma_{i-1}, \gamma_i) \right) \end{aligned} \quad (4.3.2)$$

$$\cdot \mathbb{E} \left[|G_z^\Lambda(x, x)|^s \prod_{i=1}^N |G_z^{\Lambda_i}(\gamma_i, \gamma_i)|^s \cdot |G_z^{\Lambda_{N+1}}(\gamma_{N+1}, y)|^s \right], \quad (4.3.3)$$

The resolvent $G_z^{\Lambda_j}(w, w')$ does not depend on the random variables ω_{γ_i} for $i = 0, \dots, j-1$. Hence, recursive applications of the a priori bound 3.1.4 yield

$$\mathbb{E} \left[|G_z^\Lambda(x, x)|^s \prod_{i=1}^n |G_z^{\Lambda_i}(\gamma_i, \gamma_i)|^s \right] \leq \left(\frac{1}{1-s} \frac{1}{\lambda^s} \right)^{n+1}$$

and, utilizing the deterministic bound $|G_z| \leq |\operatorname{Im}(z)|^{-1}$ (see 3.1.2) for the final step

$$\begin{aligned} & \mathbb{E} \left[|G_z^\Lambda(x, x)|^s \prod_{i=1}^N |G_z^{\Lambda_i}(\gamma_i, \gamma_i)|^s \cdot |G_z^{\Lambda_{N+1}}(\gamma_{N+1}, y)|^s \right] \\ & \leq \left(\frac{1}{1-s} \frac{1}{\lambda^s} \right)^{N+1} \mathbb{E} \left[|G_z^{\Lambda_{N+1}}(\gamma_{N+1}, y)|^s \right] \\ & \leq \left(\frac{1}{1-s} \frac{1}{\lambda^s} \right)^{N+1} \frac{1}{|\operatorname{Im}(z)|^s}. \end{aligned}$$

Inserting these estimates into the sums above, we get:

$$\mathbb{E} \left(|G_z^{(\Lambda)}(x, y)|^s \right) \leq \sum_{n=1}^N \beta(s)^{n+1} c_n^D(x, y) + R(N) \frac{1}{|\operatorname{Im}(z)|^s} \quad (4.3.4)$$

where we set $\beta(s) := \frac{1}{1-s} \frac{1}{\lambda^s}$.

Let us consider the susceptibility $\chi^D(\beta)$ of these long-range self-avoiding walks

$$\chi^D(\beta) = \sum_{w \in \mathbb{Z}^d} \sum_{N=0}^{\infty} \beta(s)^N c_N^D(x, w) = \sum_{N=0}^{\infty} c_N^D \beta(s)^N, \quad (4.3.5)$$

This power series converges absolutely if and only if $0 \leq \beta(s) < R_{\chi^D}$. Thus, for $\beta(s) = \frac{1}{(1-s)\lambda^s} < R_{\chi^D}$, the susceptibility $\chi^D(\beta)$ converges uniformly, which implies

$$R(N) \longrightarrow 0 \quad \text{as} \quad N \rightarrow \infty.$$

Sending $N \rightarrow \infty$ in (4.3.4), we obtain:

$$\mathbb{E} \left(|G_z^{(\Lambda)}(x, y)|^s \right) \leq \sum_{n=1}^{\infty} \beta(s)^{n+1} c_n^D(x, y). \quad (4.3.6)$$

Taking the supremum over all finite volumes $\Lambda \subset \mathbb{Z}^d$ now yields the two-point correlation function $C_{\beta(s)}^D(x - y)$ of the SAW:

$$\sup_{\Lambda \subset \mathbb{Z}^d} \mathbb{E} \left(|G_z^{(\Lambda)}(x, y)|^s \right) \leq \beta(s) C_{\beta(s)}^D(x - y). \quad (4.3)$$

□

Remark 4.3.2. The two theorems 3.2.1 and 3.2.2 are a consequence of this

result, combined with the decay of the two point function, proved in Section 4.1.

Remark 4.3.3. It remains to establish for which λ for $\alpha = 1$ we may find $s \in (0, 1)$ with $\beta(s) < 1/\mu_d$. As above, the unique critical point is $s_{\text{crit}} = 1 - 1/\ln \lambda$ and

$$\min_{s \in (0, 1)} \beta(s) = \beta(s_{\text{crit}}) = \frac{e \ln \lambda}{\lambda}$$

which is less than $1/\mu_d$ if and only if $\lambda > \lambda_0$ with $\lambda_0 = \mu_d e \ln(\lambda_0)$.

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